

It Is Raining Space Dust

SCUDEM X 2025 - Problem A

Team 1038

Iryna Chebitko

V. N. Karazin Kharkiv National University,
Kharkiv, UKRAINE

Sofiia Shypenko

V. N. Karazin Kharkiv National University,
Kharkiv, UKRAINE

Oleksii Vorotyntsev

V. N. Karazin Kharkiv National University,
Kharkiv, UKRAINE

Coach: Nataliya Goncharuk, Texas A&M University, College Station TX USA

V. N. Karazin KhNU

Kharkiv, UKRAINE

Outline

- 1 Introduction and Problem Statement
- 2 Mathematical Model Development
- 3 Numerical Solution Approach
- 4 Results and Analysis
- 5 Surface Distribution Estimation

The Cosmic Dust Phenomenon

- Cosmic dust constantly falls on Earth's atmosphere
- Size range: $1\mu m$ to $1000\mu m$
- Sources: comets, asteroids, and objects within the solar system
- Wide variety of shapes, sizes, and velocities

Scientific Importance

Cosmic dust provides insights into matter distribution, solar system formation, and atmospheric processes

Problem Statement

Question 1: In what ways does atmospheric entry influence the physical and dynamical properties of cosmic dust particles?

Question 2: Is it possible to reconstruct the initial distribution of cosmic particles entering Earth's atmosphere based on the characteristics of those found on the surface?

Our Goals:

- 1 Model particle behavior during atmospheric entry
- 2 Predict outcomes for different particle sizes and velocities
- 3 Estimate particle distribution at Earth's surface
- 4 Work backwards to determine initial atmospheric distribution

Forces Acting on Particles:

- Gravitational force
- Atmospheric drag
- Buoyancy (negligible)

Environmental Effects:

- Atmospheric density variation
- Temperature changes (is not modeled in our solution)
- Ablation/vaporization
- Fragmentation (negligible)

Assumptions

- **A1:** Particles are spherical (simplifies drag calculations)
- **A2:** Earth's atmosphere is stratified with exponential density model
- **A3:** Simplified ablation (no temperature tracking)
- **A4:** No wind effects initially considered
- **A5:** Constant particle density (material properties)
- **A6:** Constant drag coefficient
- **A7:** No fragmentation

Differential Equations of Motion

Initial Conditions:

$$\theta (^{\circ}), \quad v_0 \text{ (m/s)}, \quad v_{x0} = v_0 \cos \theta, \quad v_{y0} = -v_0 \sin \theta.$$

The motion of a meteoroid during atmospheric entry is described by the coupled system:

$$\frac{dx}{dt} = v_x, \quad \frac{dy}{dt} = v_y, \quad (1)$$

$$\frac{dv_x}{dt} = -\frac{\rho(h)C_d A v v_x}{2m}, \quad \frac{dv_y}{dt} = -g - \frac{\rho(h)C_d A v v_y}{2m}, \quad (2)$$

$$\frac{dm}{dt} = -\frac{1}{2}\sigma_{ab}\rho_{air}(h)Av^3 \quad (3)$$

where:

$$\rho_{air}(h) = \rho_0 e^{-h/H}, \quad A = \pi r^2, \quad r = \left(\frac{3m}{4\pi\rho_p} \right)^{1/3}, \quad v = \sqrt{v_x^2 + v_y^2}.$$

Parameter Values

Parameter	Symbol / Value	Units / Description
Gravitational acceleration	$g = 9.81$	m/s^2
Sea-level air density	$\rho_0 = 1.225$	kg/m^3
Atmospheric scale height	$H = 8500$	m
Particle density	$\rho_p = 3700$	kg/m^3 (Lost City)
Drag coefficient	$C_d = 0.47$	– (sphere)
Ablation coefficient	$\sigma_{ab} = 1.25 \cdot 10^{-8}$	s^2/m^2
Initial altitude	$h_0 = 10^5$	m (Kármán line)
Initial radius range	$r_0 = 10^{-6} - 10^{-3}$	m
Initial velocity range	$v_0 = 12 \cdot 10^3 - 72 \cdot 10^3$	m/s
Entry angle range	$\theta = 1^\circ - 90^\circ$	degrees
Simulation time span	$t = 0 - 50$	s
Integration method	RK45 / solve_ivp	–

Method Selection: Runge-Kutta 4th Order (RK45)

Justification:

- Good balance of accuracy and computational efficiency
- Stable for stiff atmospheric density gradients
- Well-suited for coupled ODE systems

Implementation Details:

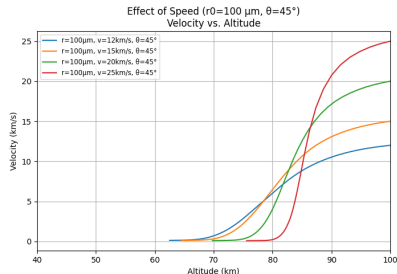
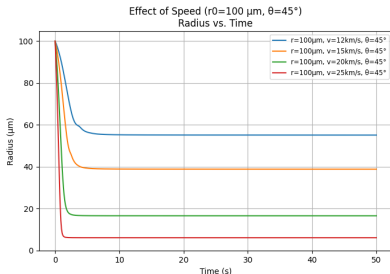
- Adaptive step size for different altitude regimes
- Termination conditions: $h \approx 0$ or $m \approx 0$ (complete ablation)

Algorithm Flowchart

Key steps:

- 1 Initialize particle parameters (θ_0, v_0, r_0)
- 2 Solve ODE system $[x, y, v_x, v_y, m]$ using RK45
- 3 Check ablation conditions at each step ($m \neq 0, y \neq 0$)
- 4 Update particle mass/radius if ablating
- 5 Record final state (survived or ablated)

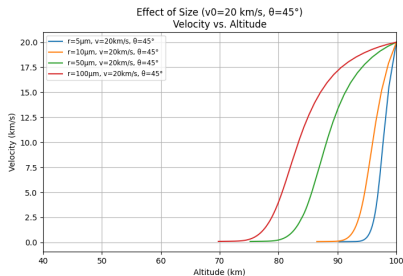
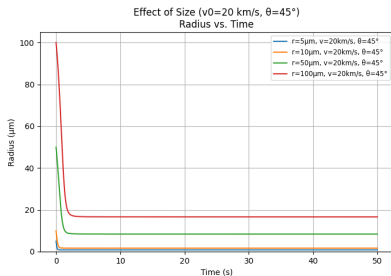
Results: Effect of Initial Velocity



Key Finding

Higher velocity is the main driver of ablation. A particle at 12 km/s survives, while the same particle at 25 km/s is almost destroyed.

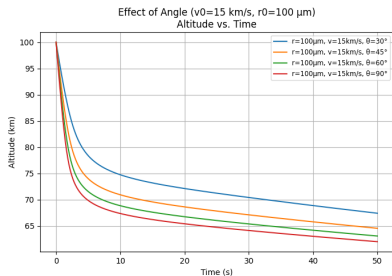
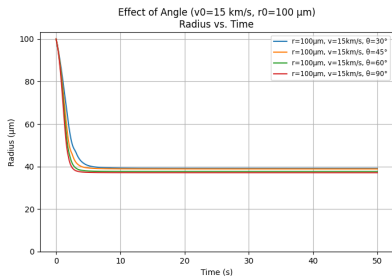
Results: Effect of Initial Size



Key Finding

Larger particles are more likely to survive. A $5\ \mu\text{m}$ particle is almost destroyed, while a $50\ \mu\text{m}$ particle penetrates deeper and survives with minimal mass loss.

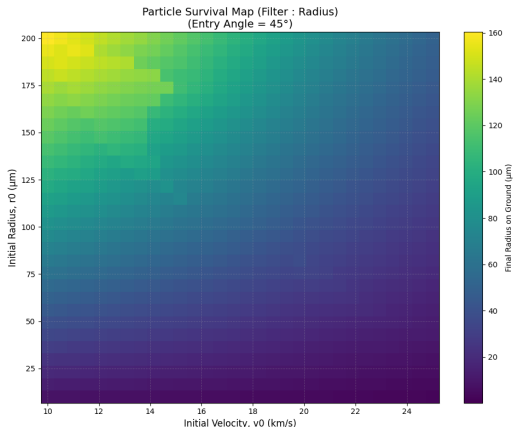
Results: Effect of Entry Angle



Key Finding

Shallow angles are more destructive. A 30° entry (“slow roast”) causes more total ablation than a steep 90° entry (“flash fry”) due to longer time spent in the atmosphere.

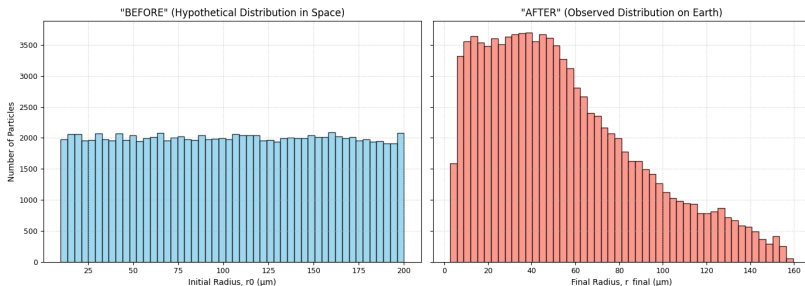
Phase Space Analysis



- **Finding 1:** High velocity and small size lead to ablation (dark colors)
- **Finding 2:** Low velocity and large size favor survival (bright colors)

Reconstructed Initial Distribution

Distortion of Particle Radius Distribution by Atmospheric Filter



Key Insights:

- **Atmospheric Filtering:** The atmosphere acts as a non-linear filter, distorting the initial particle flux.
- **Distributional Shift:** The initially uniform radius distribution becomes right-skewed upon landing.
- **Selection Bias:** Preferential ablation removes small-radius and high-velocity particles.

Answering the Problem Questions

Q1: What happens to particles during atmospheric entry?

- Our **forward model provides a transformation mapping** to predict the final particle state.
- Many particles are **irretrievably lost** due to complete ablation.

Q2: Can we work backwards to estimate the initial distribution from surface data?

Answer: Yes, with caveats

- This is an **ill-posed inverse problem** but **solvable with regularization**.
- The **accuracy depends on surface measurement quality**.

Strengths:

- Physically based model
- Captures key phenomena
- Computationally efficient

Limitations:

- Spherical particle assumption
- Simplified ablation model
- No fragmentation
- No temperature tracking

Future Work (Model Improvements):

- Incorporate non-spherical particle dynamics (e.g., rotation)
- Develop stochastic fragmentation models
- Model chemical composition effects
- Account for seasonal atmospheric variations

Conclusions & Recommendations

Overall Conclusion

The model successfully predicts particle behavior, identifies critical survival criteria, and demonstrates the feasibility of solving the inverse problem to estimate the initial atmospheric distribution from surface data.

Practical Recommendation

For a robust reconstruction, we recommend combining multiple measurement techniques (e.g., surface collection, radar observation, and satellite data).

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- ☰ Stern, E.C., White, S.M., Chen, Y.K., Arnold, J.O.. (May 17, 2017). Ablation and Heating During Atmospheric Entry and Its Effect on Airburst Risk
- ☰ Bess, T.D.. (January, 1979). Atmospheric Entry and Fragmentation of Dense Meteoroids
- ☰ Trigo-Rodríguez, J.M., Dergham, J., Gritsevich, M.. (March 27, 2021). A Numerical Approach to Study Ablation of Large Bolides: Application to Chelyabinsk

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Glory to Ukraine!